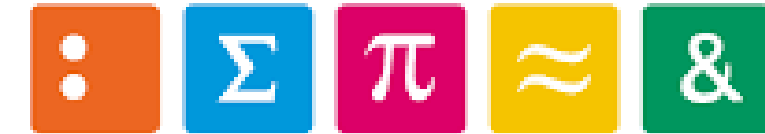




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WALLIS**

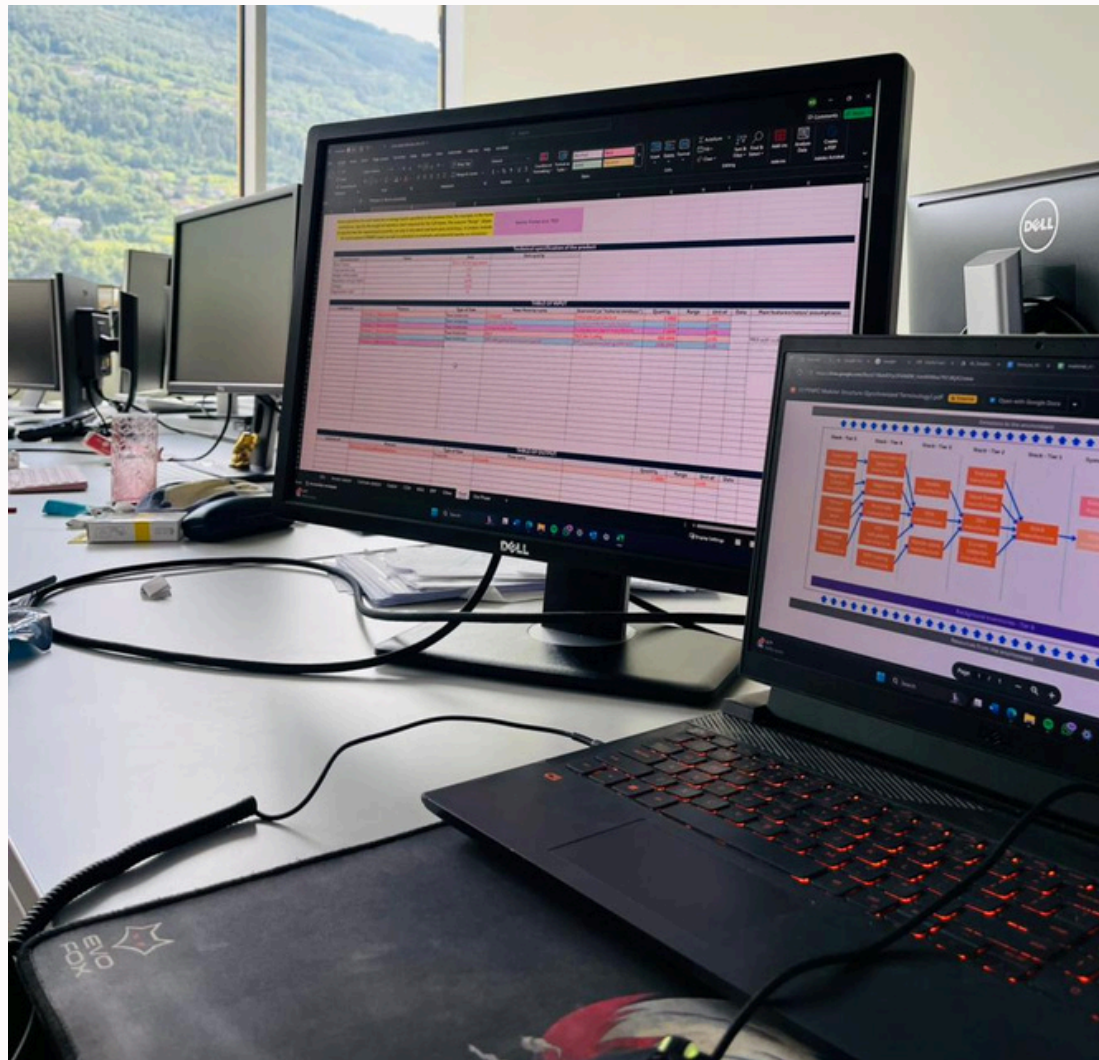


II 301 – Industrial Internship and Comprehensive Viva & II302 – Additional Internship

Presented by: Shreyas Manak

Introduction

Workplace : HES-SO Valais-Wallis, Switzerland
Department : Institute of Energy and Environment
Duration : 6 months (2nd June - 30th November)
Supervised by : Prof. Manuele Margni
Topic : LCA of a 100kW PEM Fuel Cell Stack



Research Background

Anode: $H_2(g) \longrightarrow 2H^+(aq) + 2e^-$

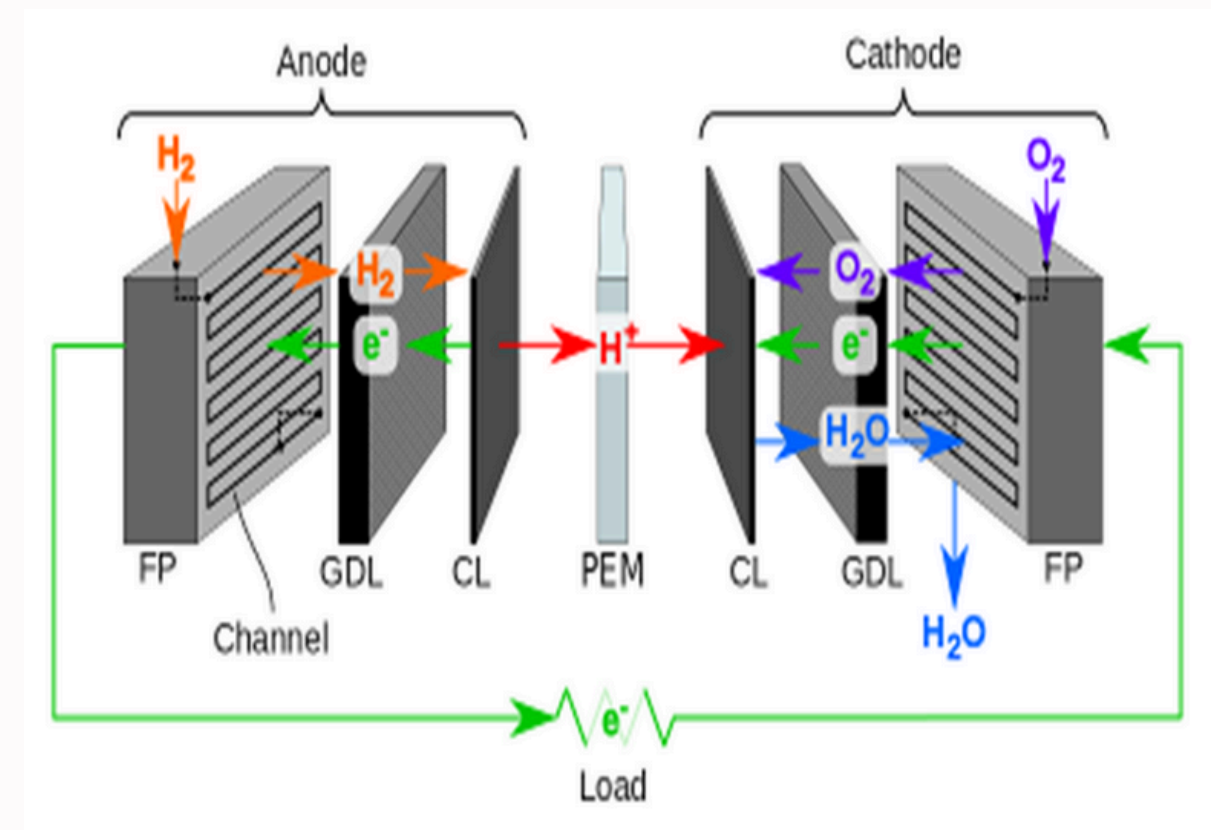
Cathode: $1/2 O_2 + 2H^+(aq) + 2e^- \longrightarrow H_2O(l)$

Overall: $H_2(g) + 1/2 O_2(g) \longrightarrow H_2O(l)$

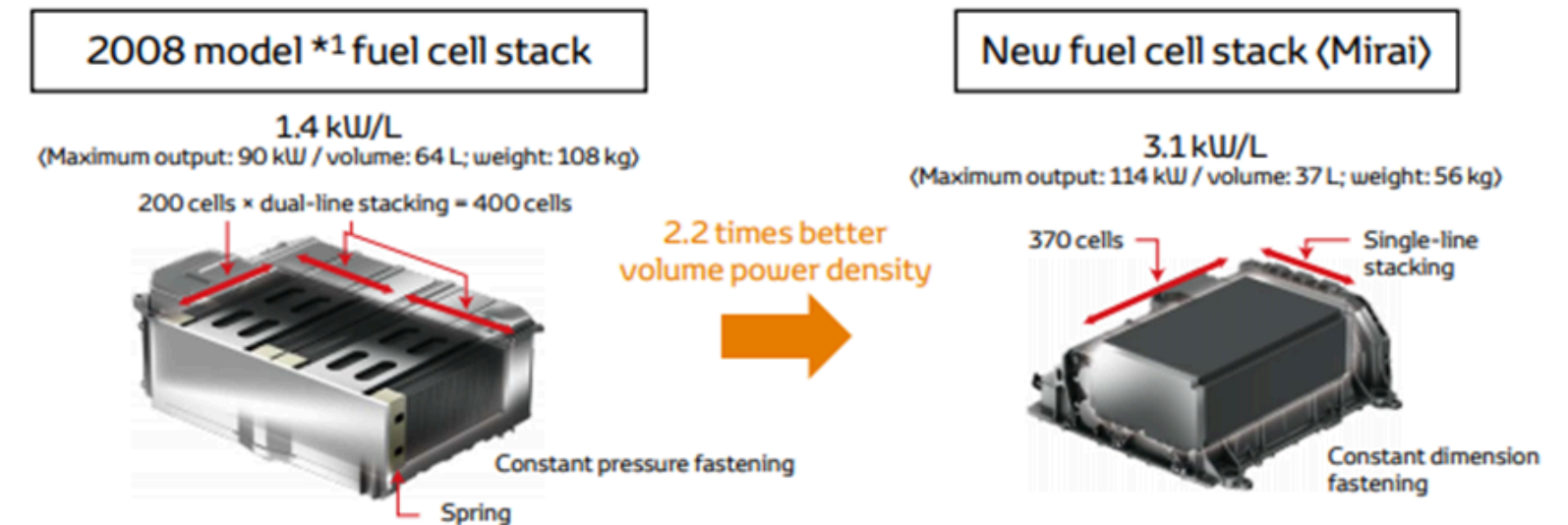
PEMFC is an electrochemical cell which converts chemical energy of H₂ and O₂ into electricity by redox reactions. It has very wide applications like in automobiles, satellites, power generation etc.

Breakthroughs in fuel cell technology have led to the creation of a smaller, lighter new fuel cell stack with enhanced performance.

By developing cheaper materials (like non-platinum catalysts) and improving system lifespan, research can fully leverage their potential for high-efficiency, zero-emission power generation.



Schematic Diagram of PEM Fuel Cell



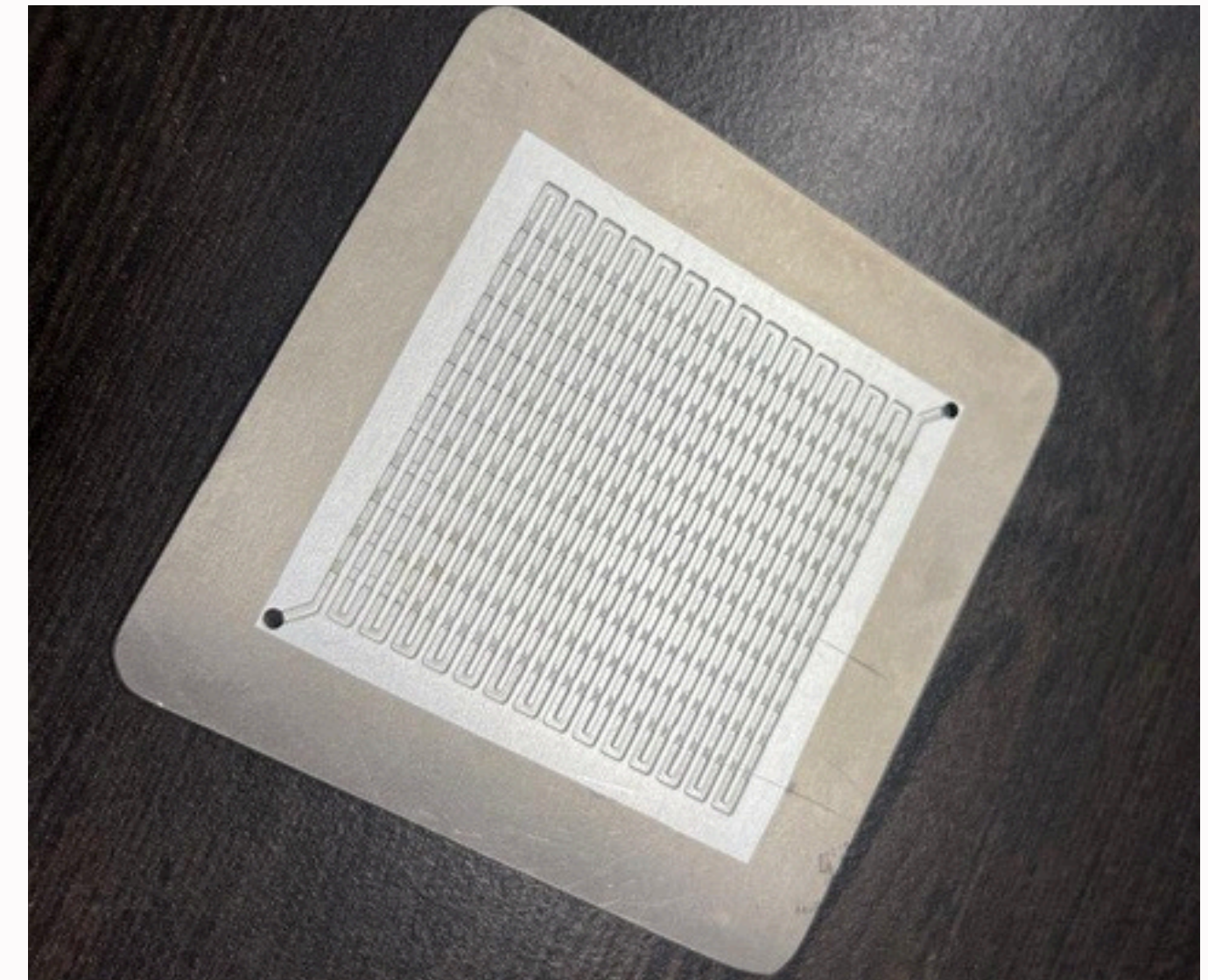
State of the art PEM Fuel Cell

Research Background

Under Prof. Dhiraj K. Mahjan at RMML lab, IIT Ropar for a DST funded project to develop a 100W PEM Fuel Cell Stack.



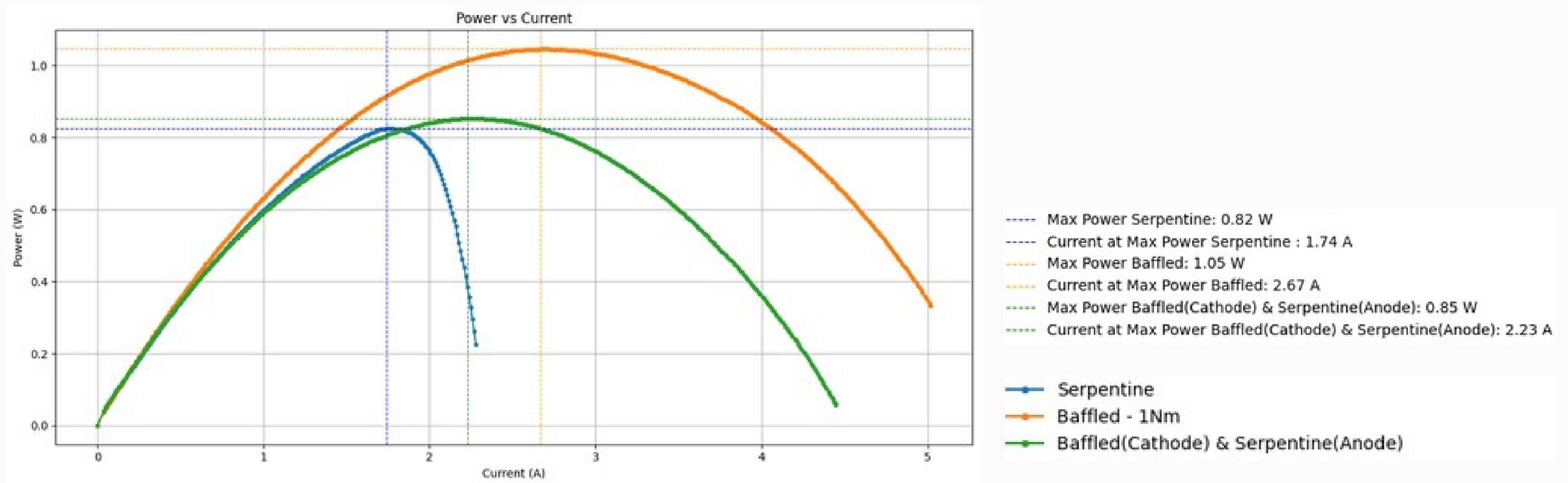
Fuel Cell Testing Rig: The setup includes a 1000W Load, Humidifier, PEM based Fuel Cell and Gas flow controllers.



Experimented with Ag coated novel baffled-serpentine titanium plates.

Research Background

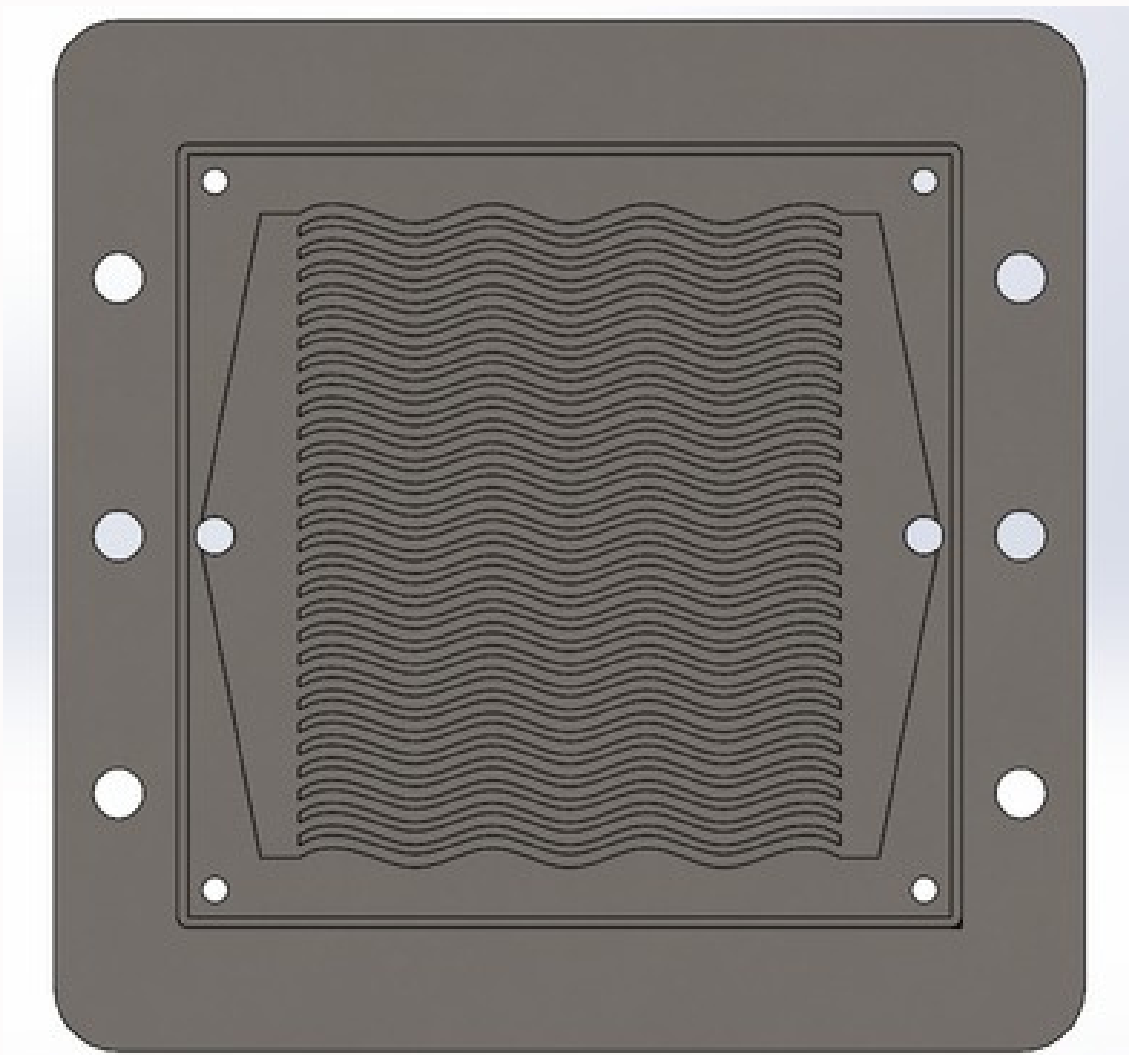
Under Prof. Dhiraj K. Mahjan at RMML lab, IIT Ropar for a DST funded project to develop a 100W PEM Fuel Cell Stack.



Compiled comparison between serpentine and baffled-serpentine plates using the data acquired through a program made for the software compatible to the load.

Research Background

Under Prof. Dhiraj K. Mahjan at RMML lab, IIT Ropar for a DST funded project to devolp a 100W PEM Fuel Cell Stack.



Developed CAD Model for the coolant side of the BPP to keep the fuel cell stack within operating temperatures.

Configuration	Clamping Torque (in N-m)	Clamping Force (in kN)	OCV (in V)	Max. Power (in W)
Serpentine at both Anode and Cathode	1.50	7.69	0.97	1.057
	1.25	6.41	0.97	1.076
	1.00	5.13	0.99	1.153
	0.90	4.61	0.98	1.221

Experimental results to find optimum clamping torque for our fuel cell.



Prof. Manuele's Lab

Academic Positions & Affiliations:

- Full Professor, Institute of Energy and Environment at HES-SO Valais-Wallis, Switzerland
- Full Professor, Department of Mathematical and Industrial Engineering at Polytechnique Montréal, Canada
- Co-chair Holder of the International Life Cycle Chair (CIRAIG)
- Academic Guest and Lecturer at EPFL

Research Focus:

- Strategic Goal: Using LCA to support strategic planning and policy development for a sustainable energy future.
 - Energy Transition Focus: Quantifying the environmental impacts of decentralized energy systems, specifically within the Swiss context.
 - Hydrogen Technologies Analysis: Comprehensive LCA to evaluate the impacts of:
 1. Different electrolyzer and fuel cell technologies.
 2. Their integration with renewable energy sources.
 - Overall Objective: Creating data-driven pathways for realizing carbon-neutral industrial and energy solutions.
- 

My Role

Life Cycle Assessment (LCA) of a 100 kW PEM Fuel Cell Stack:

- To model the stack's lifecycle using the Ecoinvent database within the brightway framework under the ECOPEM project.
- To conduct a hotspot analysis to identify primary environmental impact contributors for the eco-design of the future stack.
- The findings from this assessment will serve as a baseline for the ECOPEM project, which is funded by the European Union and Switzerland's State Secretariat for Education, Research and Innovation (SERI)'s Clean Hydrogen Partnership.



Progress

Month 1: June

- Understood LCA through Prof. Manuele's course for EPFL students.
- Implemented the learnings through practice in the Activity Browser software.
- Familiarized with the ecoinvent and biosphere3 database.
- Conducted a literature review on LCA studies of Fuel Cell and Electrolyzer technologies.

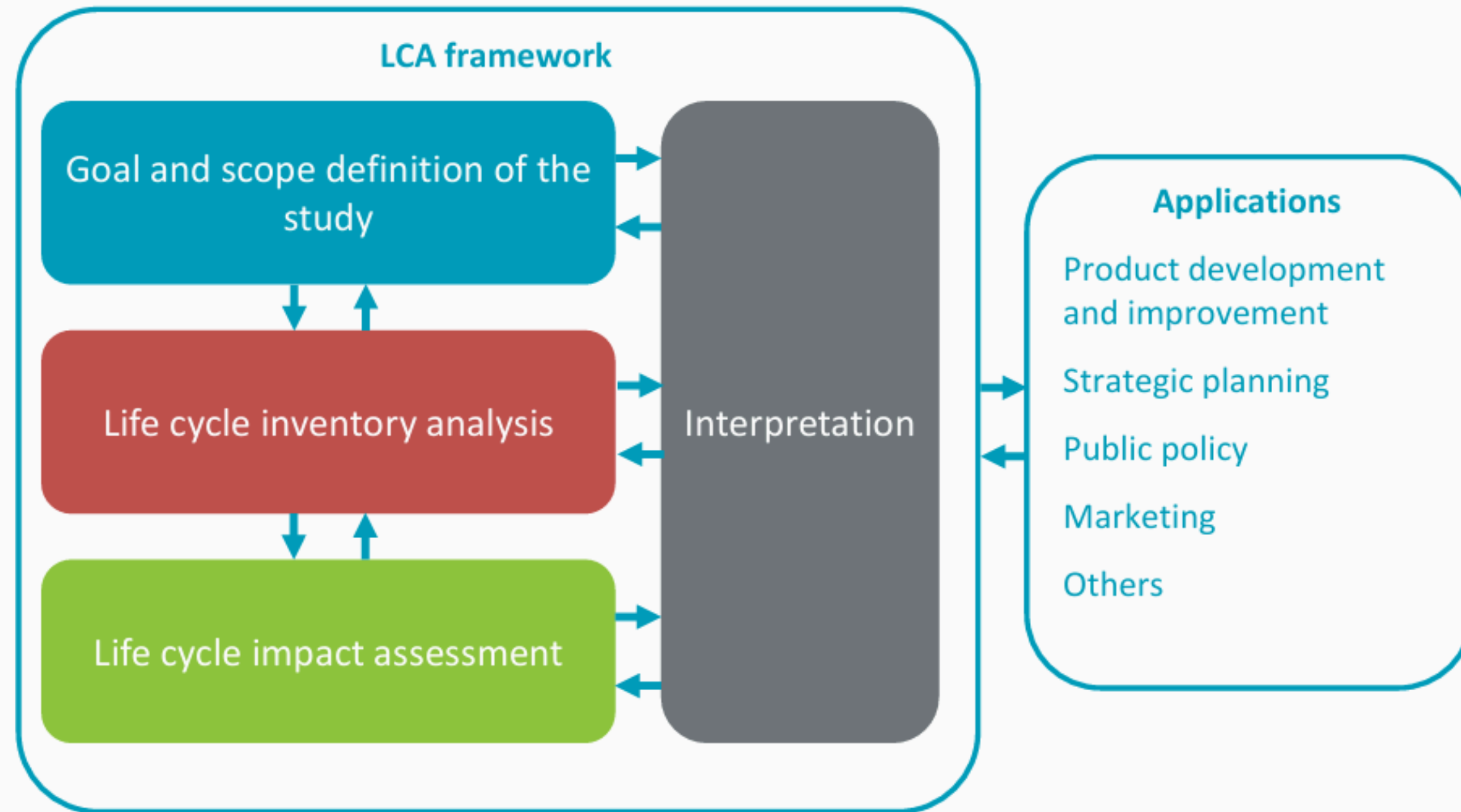
The screenshot displays the Activity Browser software interface. On the left, the 'Databases' panel shows a table with columns: Name, Records, Read-only, Depends, and Modified. The 'biosphere3' database is selected, showing 4079 records. Below this, the 'Datasets' panel for 'biosphere3' shows a table with columns: name, categories, type, unit, and modified. The table lists various chemical compounds and their associated categories and units.

On the right, the 'Welcome to the Activity Browser!' message is displayed. It includes the Activity Browser logo and a description: 'The Activity Browser is an open source graphical user interface designed to increase the productivity when working with the Brightway2 advanced life cycle assessment framework.' Below the message, the 'Key features' are listed:

- Manage brightway2 projects, databases and activities (increasing your productivity with brightway2)
- Calculate fast LCA results (use "calculation setups" to calculate LCA results for several activities and LCIA methods at once)
- Easily plot and export your LCA results (contribution analyses, Monte Carlo simulations)
- Visualize your results in Sankey diagrams or explore your database with the Graph Explorer

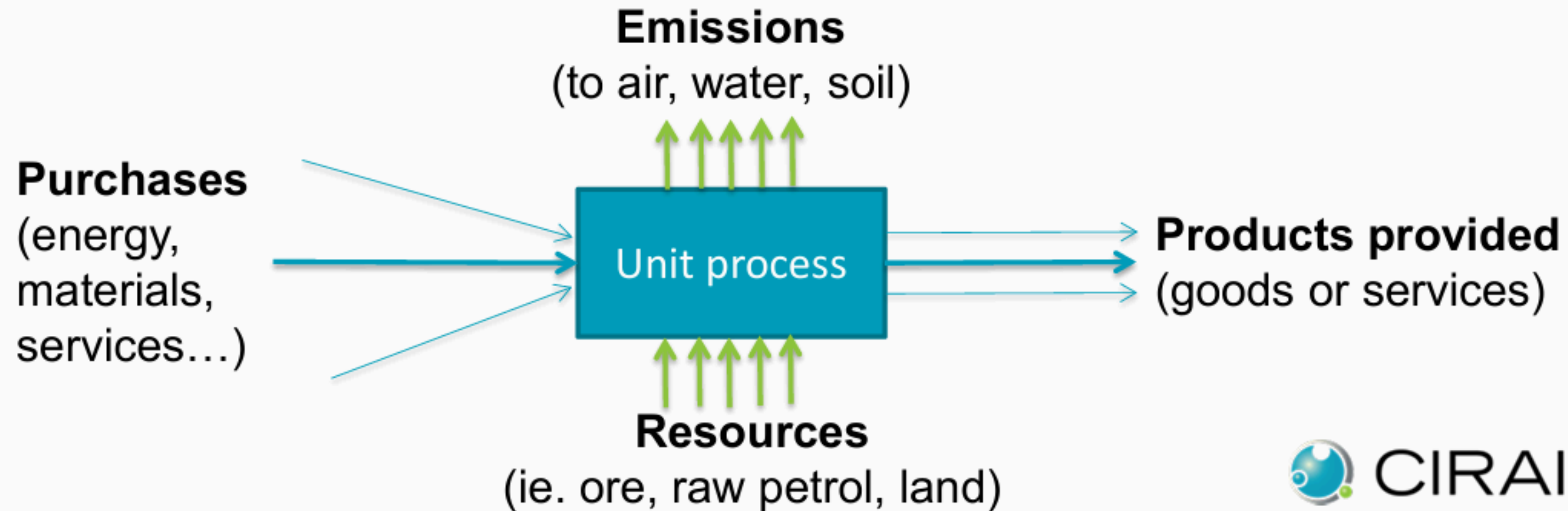
Below the key features, the 'Examples' section shows three visualizations: 'LCA results overview' (a bar chart showing LCA results for different activities), 'Monte Carlo simulation' (a histogram showing the distribution of LCA results), and 'Sankey diagrams' (a flow diagram showing the contribution of different activities to the total LCA results).

Life Cycle Assessment (LCA)



A systematic method for evaluating the potential environmental impacts of the processes involved to deliver the mentioned final demand (Functional Unit).

Unit Processes

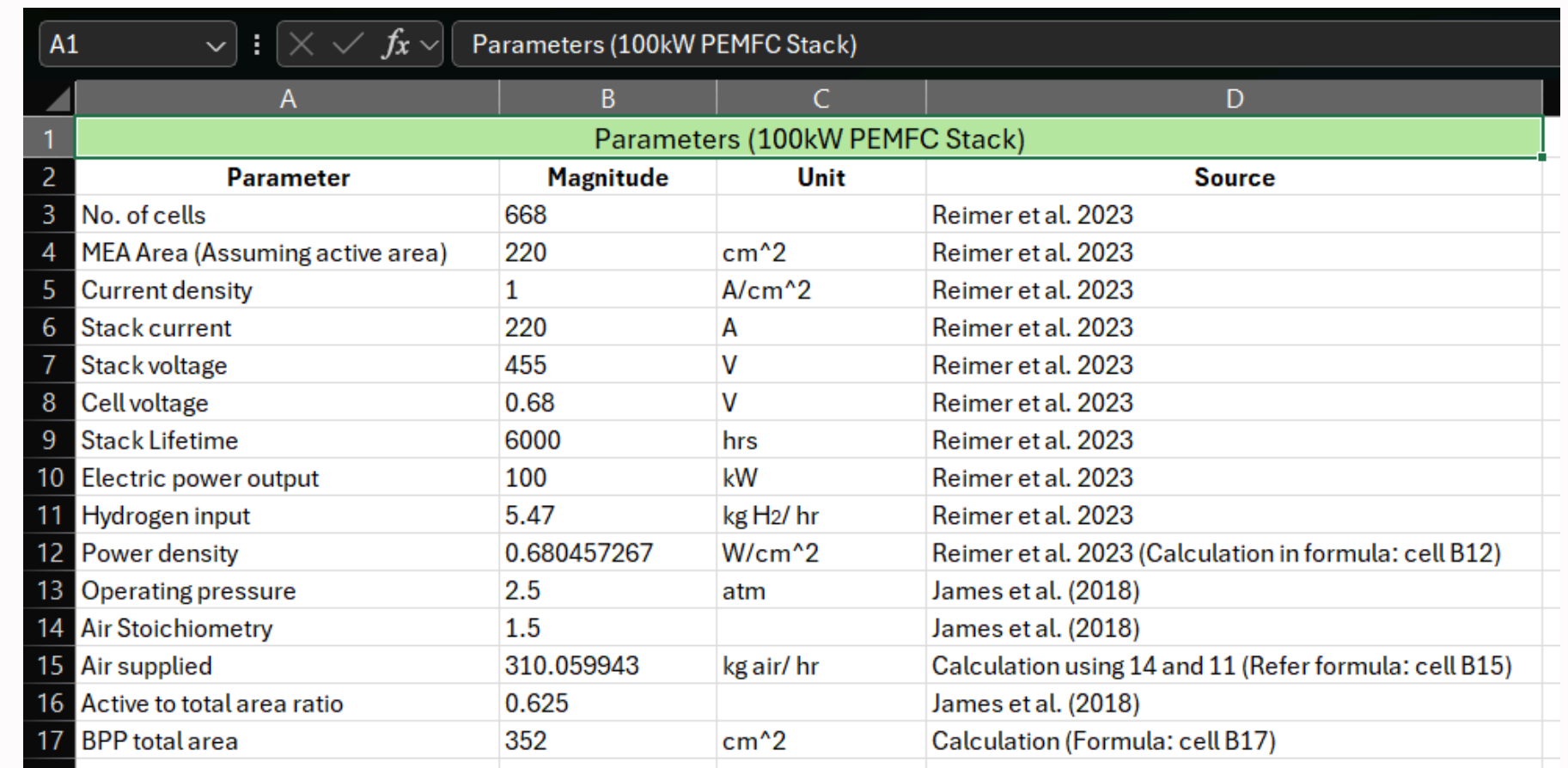


- Elementary Flows (Biosphere3 database)
- Intermediary Flows (Ecoinvent database)

Progress

Month 2: July

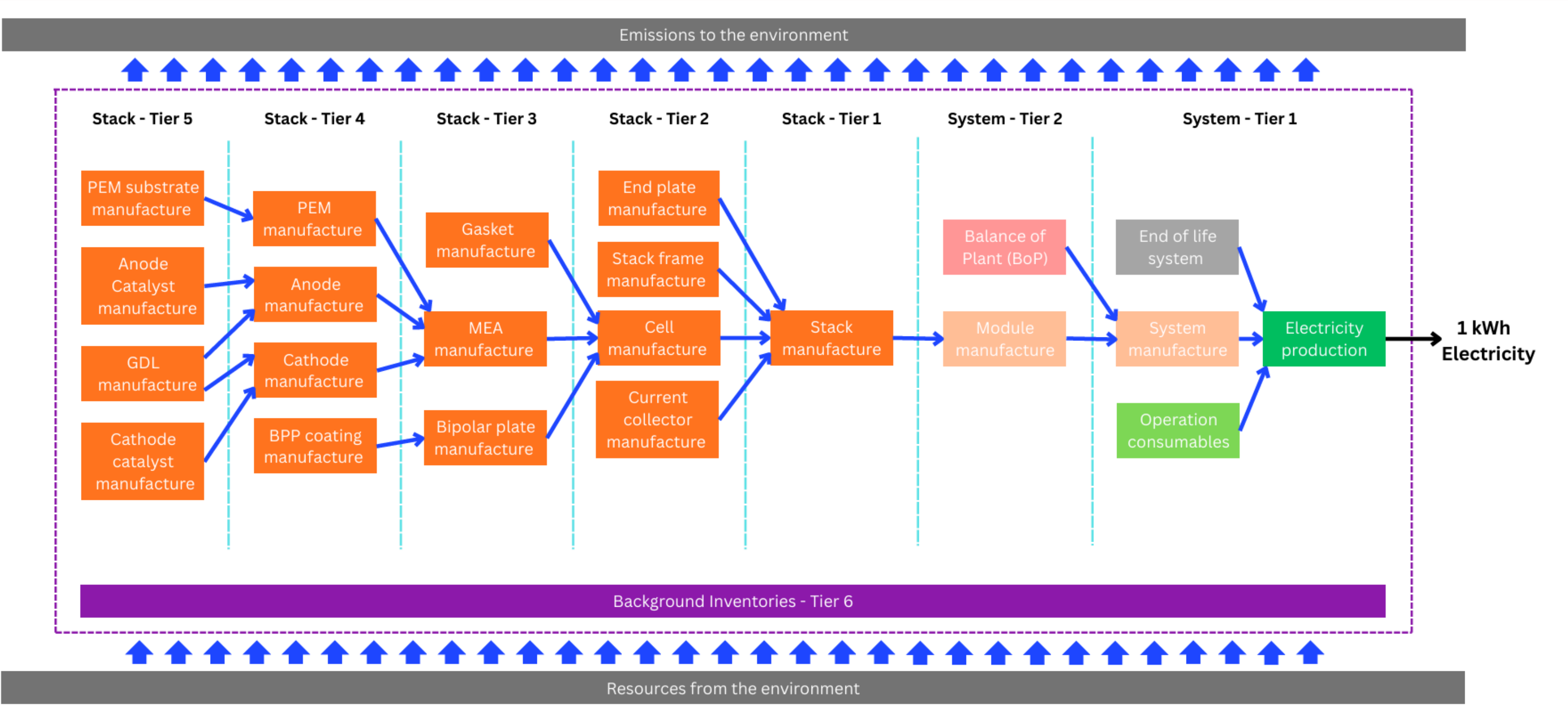
- Defined Goal and Scope of the study.
- Familiarized with the modular approach for LCA devised by HES-SO.
- Created a process tree for my study, based on the modular structure.
- Prepared the list of materials and the parameters of the stack.



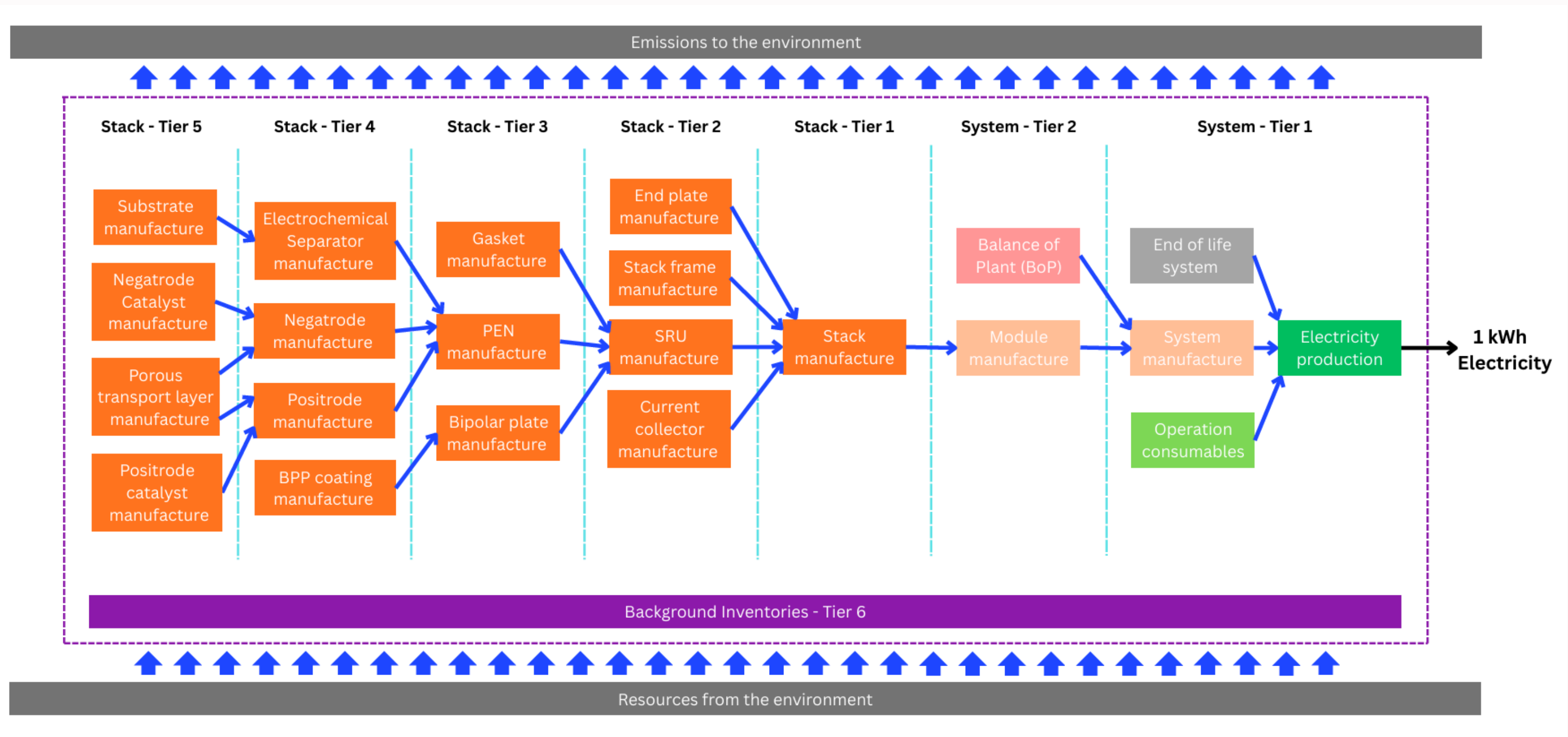
Parameters (100kW PEMFC Stack)			
Parameter	Magnitude	Unit	Source
No. of cells	668		Reimer et al. 2023
MEA Area (Assuming active area)	220	cm ²	Reimer et al. 2023
Current density	1	A/cm ²	Reimer et al. 2023
Stack current	220	A	Reimer et al. 2023
Stack voltage	455	V	Reimer et al. 2023
Cell voltage	0.68	V	Reimer et al. 2023
Stack Lifetime	6000	hrs	Reimer et al. 2023
Electric power output	100	kW	Reimer et al. 2023
Hydrogen input	5.47	kg H ₂ / hr	Reimer et al. 2023
Power density	0.680457267	W/cm ²	Reimer et al. 2023 (Calculation in formula: cell B12)
Operating pressure	2.5	atm	James et al. (2018)
Air Stoichiometry	1.5		James et al. (2018)
Air supplied	310.059943	kg air/ hr	Calculation using 14 and 11 (Refer formula: cell B15)
Active to total area ratio	0.625		James et al. (2018)
BPP total area	352	cm ²	Calculation (Formula: cell B17)

Fig: Parameters (100kW PEMFC Stack)

Modular Structure



Modular Structure (Synchronus Terminology)



Broad Modelling Steps

Process Tree

Used to identify
unit processes

**Model:
Product
Sysytem**

Used to
realize

Reality





Goal

Life Cycle Assessment (LCA) of a 100 kW PEM Fuel Cell Stack:

Primary Objective: To conduct a hotspot analysis on the PEMFC stack.

Impact Identification: To identify the primary contributors to the potential environmental impact across the overall life cycle at different tiers of the modular structure.

Benchmarking: To establish a baseline for comparison with future PEMFC stacks , specifically serving as a reference for the ECOPEM project.

Communication of Results: To communicate the main impact contributors to the technical experts of the ECOPEM project , guiding their innovation efforts toward an environmentally friendly (non-fluorinated) design.





Scope

Life Cycle Assessment (LCA) of a 100 kW PEM Fuel Cell Stack:

Functional Unit: Defined as 1 kWh of net electricity delivered from the 100 kW PEMFC stack operating within Europe during 2023.

Life Cycle Inventory (LCI) Modeling: The Ecoinvent-3.11-cutoff database is the primary source, supplemented by a material database developed by HES-SO experts.

Life Cycle Impact Assessment (LCIA): Will be performed using Impact World+ v.2.1 (footprint version).

Limitation (Cut-off): This preliminary LCA deliberately cuts-off the Balance of Plant (BoP) components due to a current lack of data. The study acknowledges this as a significant limitation, noting that the BoP is a substantial contributor to the total environmental impact. Future research will integrate these components.



Progress (II302)

Month 3: August

- Conducted literature review to collect data for Life Cycle Inventory (LCI).
- Made calculations to filter and adjust the data according to the functional unit through the support of certain assumptions.
- Started implementing the LCI in the Activity Browser according to the modular structure.

Fig: Data collection Excel for LCI

Month 3: August

- Conducted literature review to collect data for Life Cycle Inventory (LCI).
- Made calculations to filter and adjust the data according to the functional unit through the support of certain assumptions.
- Started implementing the LCI in the Activity Browser according to the modular structure.

[illegible]

Fig: Data collection Excel for LCI

Life Cycle Inventory (LCI)

1. Obtain data for each unit process of the product-system. (scaled to one product unit)
2. Scale the intermediary flows to the functional unit.
3. Scale the elementary flows to each of the unit processes to the functional unit.
4. Sum all the elementary flows on the entire life cycle

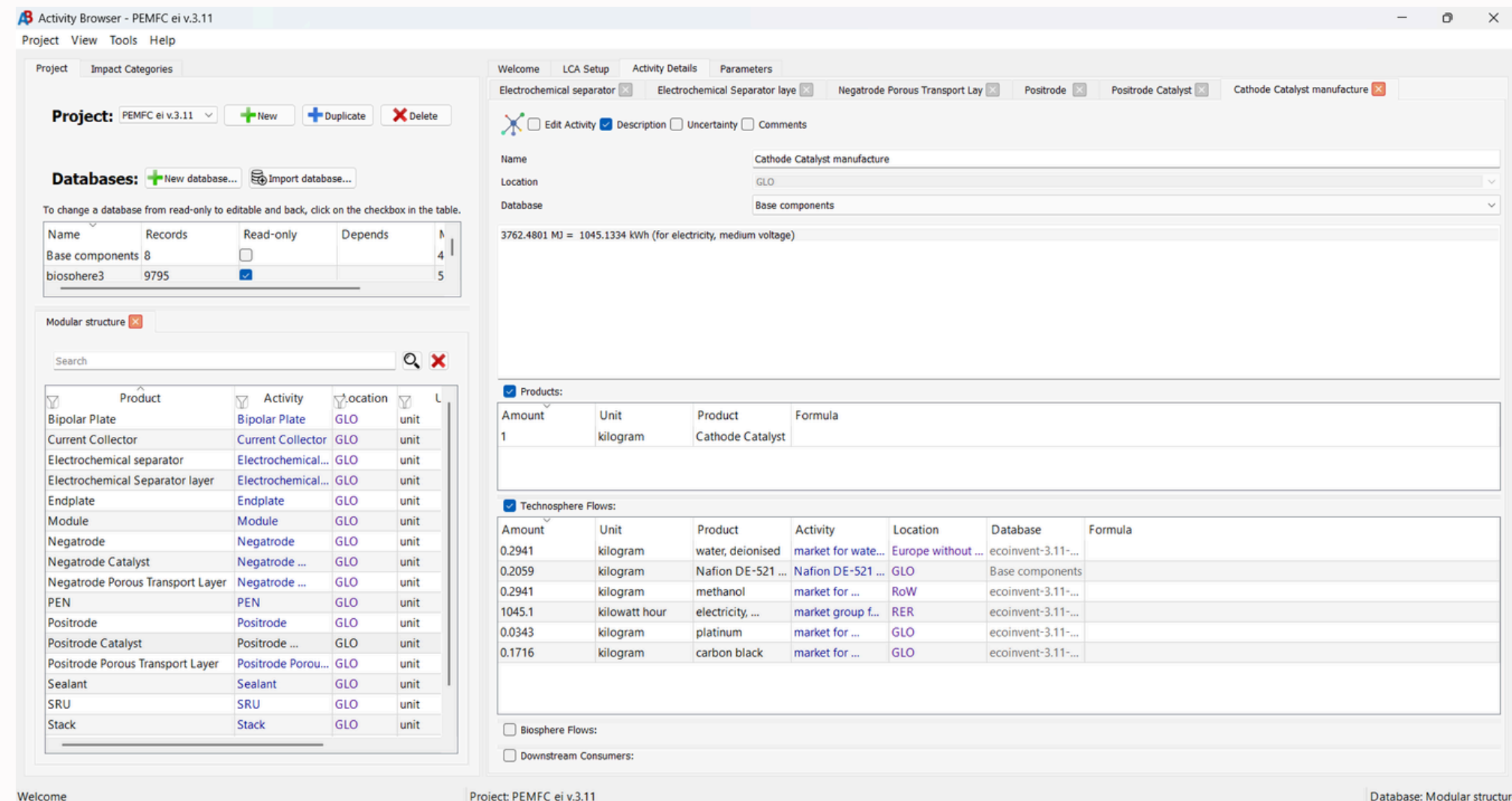


Fig: LCI implemented in activity browser.



Future Work

Life Cycle Assessment (LCA) of a 100 kW PEM Fuel Cell Stack:

Life Cycle Inventory(LCI): Complete the on-going LCI in the Activity-Browser in accordance to the modular structure.

Life Cycle Impact Assessment (LCIA): To identify high impact processes and flows (hotspots), and interpret these findings to inform improved eco-design of the stack.

Presentation: Present the interpretation of the results to the LCA experts in lab for feedback and improvements.

Sensitivity Analysis: Varying key data inputs or assumptions to determine how significantly they influence the final environmental impact results. This is done to verify the robustness of the study's conclusions and prioritize data quality for sensitive parameters.

Address Limitation: Collect data to integrate the Balance of Plant (BoP) components into the final study, thereby mitigating the major limitation and providing a more comprehensive environmental assessment.



Life Cycle Impact Assessment (LCIA)

IMPACT CATEGORIES

Environmental
Interventions (LCI)

Midpoint level

Damage level

Damage level
(aggregated into
areas of protection)

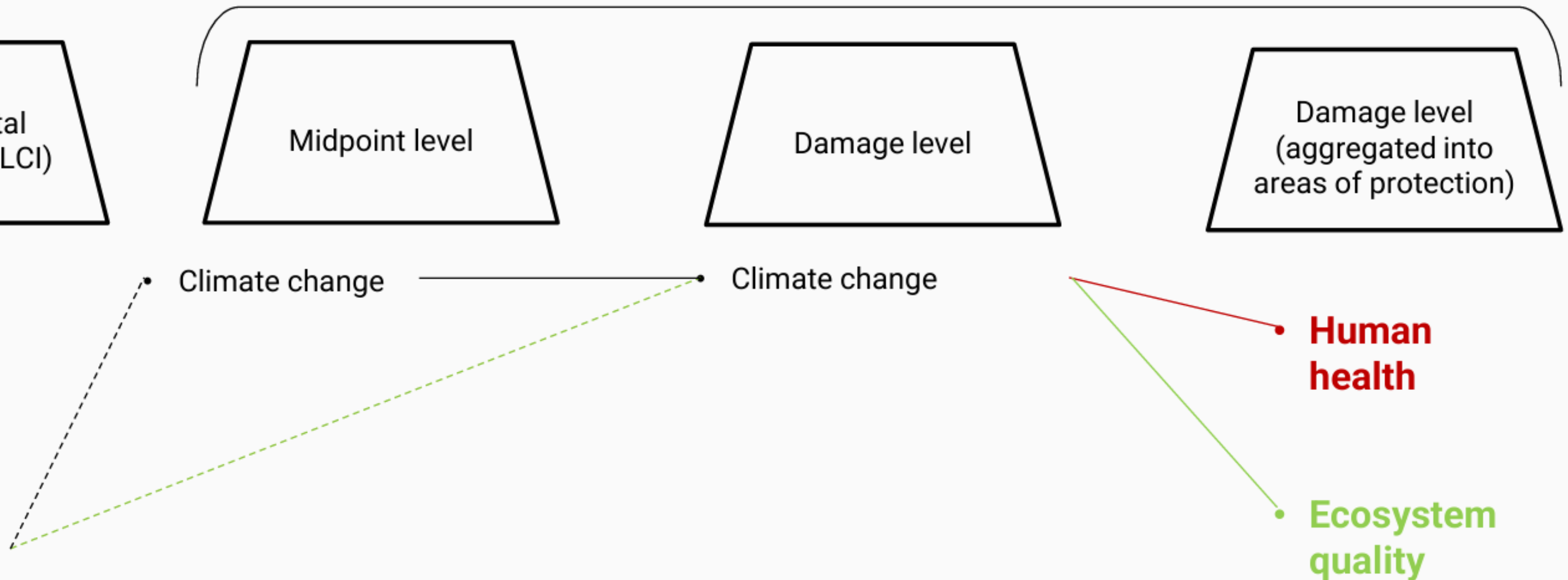
Climate change

Climate change

• **Human
health**

• **Ecosystem
quality**

Elementary
flows

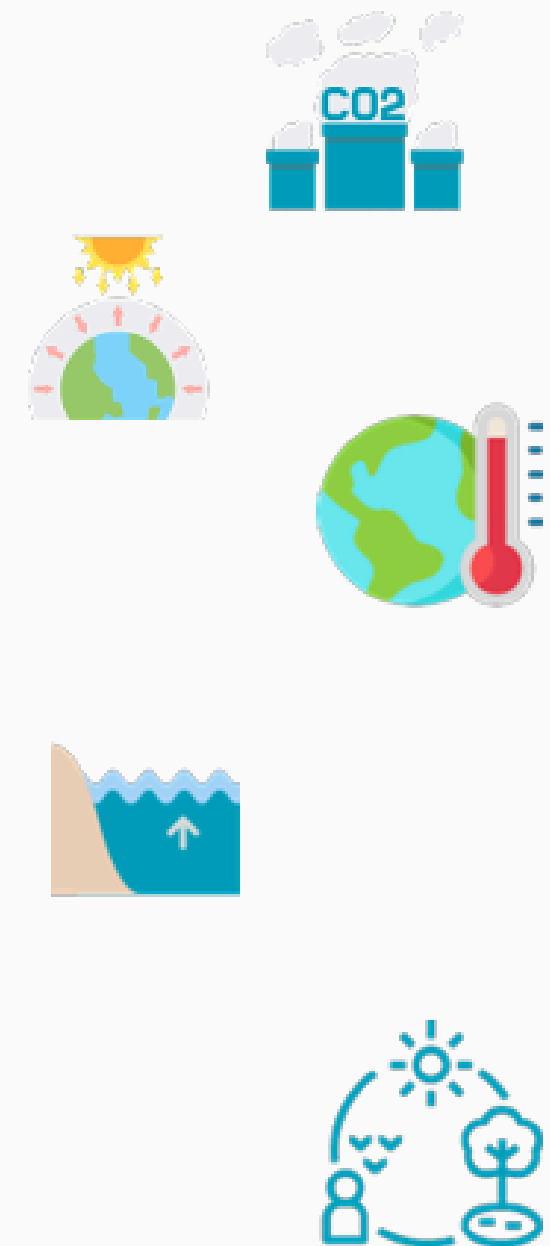


Terminologies

Term	Example
Impact category	Climate change
Inventory results	Quantity of a greenhouse gas per functional unit
Characterisation model	Model created by IPCC (International Panel on Climate Change)
Category indicator	Infrared radiative forcing (W/m ²)
Characterisation factor	Global warming potential per greenhouse gas (kg equivalent CO ₂ /kg of gas) (GWP)
Category indicator result or impact score	kg equivalent CO ₂ per functional unit
Final impact per category	Human health, Ecosystem health

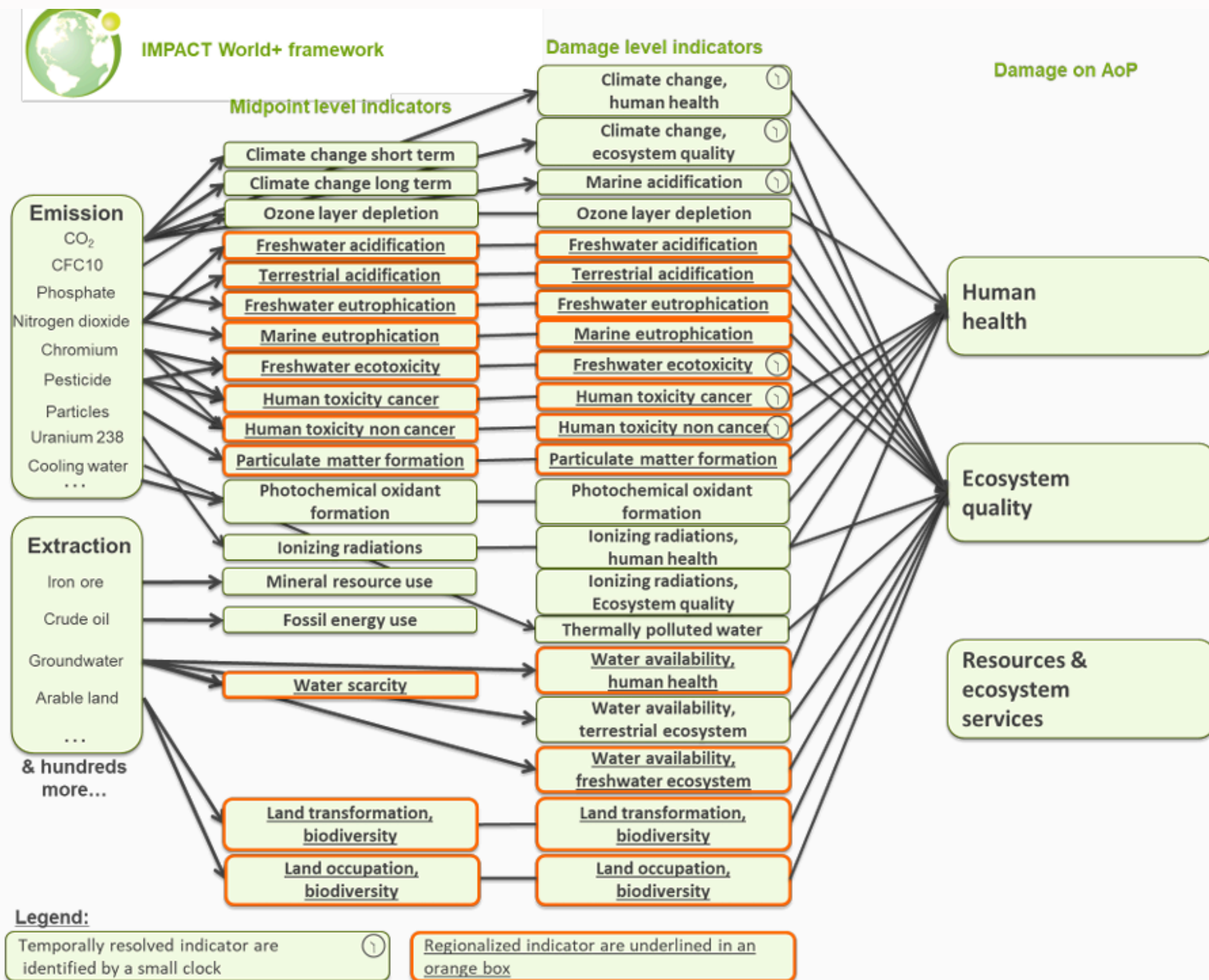
Cause-Effect Chain

1. GHG emissions (inventory results)
2. Disruption of the radiation balance
3. Global temperature rise
4. Change in primary production, change in biomes (desertification, species migration, melting ice, forest fires, ...); malnutrition, extreme events, etc.
5. Damage to ecosystems and human health



Impact World+ Framework

- 18 problem-oriented (midpoint) categories
- 21 damage-oriented categories
- 3 areas of protection



Damage oriented Impact categories and Indicators - IMPACTWorld+

Damage impact categories	Unit
Land occupation, biodiversity	PDF.m2.yr
Land transformation, biodiversity	PDF.m2.yr
Freshwater acidification	PDF.m2.yr
Marine acidification	PDF.m2.yr
Terrestrial acidification	PDF.m2.yr
Freshwater eutrophication	PDF.m2.yr
Thermally polluted water	PDF.m2.yr
Water availability, human health	DALY
Water availability, terrestrial ecosystem	PDF.m2.yr
Water availability, freshwater ecosystem	PDF.m2.yr
Marine eutrophication	PDF.m2.yr
Mineral resources use	DALY
Climate change, human health	DALY
Climate change, ecosystem quality	PDF.m2.yr
Respiratory effects	DALY
Ozone Layer Depletion	DALY
Human toxicity cancer	DALY
Human toxicity non cancer	DALY
Freshwater ecotoxicity	PDF.m2.yr
Ionizing radiations, human health	DALY
Ionizing radiations, ecosystem quality	PDF.m2.yr

Area of Protection	Unit
Human health	DALY
Ecosystem Quality	PDF.m ² .yr
Resources and ecosystem services	\$

DALY or "disability adjusted life years": Number of years (in good health) lost, premature mortality or a certain degree of illness for a certain time.

PDF.m².years or "potential of disappeared fraction of species": Percentage of species disappeared over a certain surface over a certain amount of time.



**Thank
You**